

Decision Log — PS 26083: Extreme Heatwave Early Warning & Human Thermal Stress Index

Every non-trivial project decision, in order. Format: Decision / Why / Source / Prototype assumption / Future calibration.

D1 — City selection: Ahmedabad

Decision: Build the prototype for Ahmedabad, not a generic/national system.

Why: Data availability beats city glamour. Ahmedabad has (a) a well-known, well-documented Heat Action Plan, (b) publicly available ward boundary GeoJSON/Shapefile (48 wards), (c) full ERA5 coverage, (d) a strong recorded heatwave history (2010 Ahmedabad heatwave is a landmark public-health case study in India).

Source: AMC official info (7 zones, 48 wards); OpenCity/DataMeet ward boundary datasets.

Prototype assumption: Single-city prototype; not yet generalized to other cities.

Future calibration: Architecture is designed to be city-agnostic — swapping ward boundaries + demographic source should port the pipeline to any Indian city with similar data availability.

D2 — Weather data source: ERA5 (not IMD/other)

Decision: Use ECMWF ERA5 reanalysis (hourly, single-level) as the weather backbone.

Why: Free, global, hourly resolution, includes temperature, dewpoint, 10m U/V wind, and radiation variables needed for MRT/UTCI — all in one consistent, citable dataset. CDS API access is straightforward for a 7-day build.

Source: Copernicus Climate Data Store (CDS), ERA5 hourly single-level dataset.

Prototype assumption: ERA5 is REANALYSIS, not a live forecast. We do not call it a "forecast" anywhere in the project — this is a historical-replay prototype, clearly labeled.

Future calibration: Phase 2 would swap in a genuine 3–5 day forecast product (e.g. NCMRWF/IMD operational forecast) feeding the same UTCI→NCTL→CTBI pipeline.

D3 — Thermal stress metric: UTCI (not just Heat Index or WBGT alone)

Decision: UTCI (Universal Thermal Climate Index) is the primary human thermal stress metric. WBGT is a secondary/comparison metric only.

Why: UTCI integrates air temperature, humidity, wind speed, AND mean radiant temperature (solar/thermal radiation) — the PS explicitly calls out that dry-bulb-only warnings miss humidity/wind/radiation compounding effects. UTCI is the most complete of the standard indices for this. WBGT requires a separate natural-wet-bulb/black-globe derivation path that would consume real build time without adding much beyond what UTCI already captures.

Source: `pythermalcomfort` library's official `utci()` implementation (industry-standard, peer-reviewed).

Prototype assumption: Using the library's established formula, not a from-scratch re-derivation — this is standard practice, not a shortcut.

Future calibration: None needed — UTCI itself doesn't need recalibration; only the surrounding NCTL/CTBI framework does.

D4 — MRT (Mean Radiant Temperature) derivation from ERA5 radiation variables

Decision: Compute MRT from ERA5's surface solar/thermal radiation downwards and related radiative variables (fdir, ssrd, ssr, strd, str), following the established ERA5-HEAT/thermofeel radiation-to-MRT method (regrid, convert accumulation period, compute lup/Rsw/Dsw/Uw radiation components, then MRT via Stefan-Boltzmann-based projected-area-factor formulation).

Why: This is a documented, established method (used by ERA5-HEAT and other operational heat-stress tools), not an invented shortcut. It's the correct hard technical dependency that gates the entire UTCI pipeline.

Source: ERA5-HEAT / thermofeel methodology; implemented in `src/mrt.py`, `src/cos_solar.py`.

Prototype assumption: Single global radiation grid at ERA5's native 0.25°/0.1° resolution — not corrected for local shading/building geometry (an urban-canyon effect UTCI at this scale can't capture).

Future calibration: Higher-resolution radiation data (satellite-derived, building-aware) would refine MRT locally.

D5 — NCTL (Nighttime Cumulative Thermal Load) — the project's core innovation

Decision:

- Night window: 20:00–05:59 IST (used consistently in the production `daystress_and_baseline.py` — `NIGHT_START=20`, `NIGHT_END=6`).
- Nighttime baseline = 90th percentile of all historical nighttime UTCI values in the dataset (not an arbitrary fixed number).
- $NCTL(t) = \sum \max(0, UTCI_{night}(h) - UTCI_{baseline}) \times \Delta h$

Why: Standard heatwave alerts miss persistent nighttime heat, which prevents physiological recovery and is strongly linked in heat-mortality literature to compounding

health risk over multi-day heatwaves. This is the project's differentiator — "operationalizing nighttime thermal load" as a named, computable metric.

Source: Heat-health literature on nighttime recovery/non-recovery during heatwaves; baseline method is self-derived from the dataset (documented, reproducible).

⚠ Known minor inconsistency (documented, not blocking): The notebook's NCTL night-window computation (hour<8, i.e. 20:00–07:59, 12 hours) does not exactly match `daystress_and_baseline.py`'s baseline night-window (20:00–05:59, 10 hours) — a 2-hour (06:00–08:00) mismatch. Flagged during Dev's Day 3 code review. Does not change the qualitative CTBI pattern (confirmed by Khushi's Day 3 sanity check, $\text{NCTL} \leftrightarrow \text{peak-CTBI}$ correlation = 0.944 across all 6 seasons) but should be reconciled before final submission for full internal consistency.

Future calibration: With real hospitalization data, the baseline percentile threshold (90th) itself could be tuned against actual health outcomes.

D6 — CTBI (Cumulative Thermal Burden Index) formula and α sensitivity

Decision: $\text{CTBI}(t) = \alpha \cdot \text{CTBI}(t-1) + \beta \cdot \text{DayStress_norm}(t) + \gamma \cdot \text{NCTL_norm}(t)$, with α tested at 0.60/0.70/0.80, $\beta=\gamma=0.15$, $\alpha=0.70$ locked as the default.

Why: A first-order recurrence captures multi-day heat memory (today's burden depends partly on yesterday's), which single-day metrics miss. $\alpha=0.70$ was chosen as the default because it's the only tested value where $\alpha+\beta+\gamma$ sums to exactly 1.0 (see D6a below) — a mathematically clean weighted average.

Source: Standard exponential-memory/recurrence pattern used in cumulative-stress indices generally; weights chosen and locked by the team, not from an external paper (explicitly NOT claimed as "the scientifically correct decay factor" — see project's "cannot claim" list).

Prototype assumption: Linear recurrence with fixed weights; not fit against real health outcome data (none available).

Future calibration: Once real hospitalization/mortality data exists, $\alpha/\beta/\gamma$ could be fit via regression against actual outcomes instead of chosen for mathematical cleanliness.

D6a — CTBI weight-sum audit finding (Shweta, Day 3)

Finding: $\alpha+\beta+\gamma = 1.0$ only at $\alpha=0.70$ (the locked default). At $\alpha=0.60$, weights sum to 0.90 (slight downward drift at steady state); at $\alpha=0.80$, they sum to 1.10 (slight upward drift). This does NOT invalidate the sensitivity test — the playbook only requires the *qualitative pattern* (rise-to-peak, decline-after) to remain stable across α , which independently-verified sanity checks confirm it does. But $\alpha=0.6/0.8$ CTBI values are not on an identical numeric scale to $\alpha=0.7$ — worth stating plainly in Q&A rather than implying all three runs are directly comparable in magnitude.

Verified by: docs/Shweta_Day3_CTBI_Normalization_Audit.md — full min-max bounds check also confirmed correct (DayStress_norm, NCTL_norm both properly bounded [0,1], hit true extremes, divide-by-zero guarded).

D7 — Vulnerability data: 2011 Census REJECTED due to ward-boundary mismatch

Decision: Do NOT use the 2011 Census ward-level Primary Census Abstract (DDW_PCA2407_2011_MDDS) to populate demographic_join_template.csv, despite it containing real, downloadable, ward-level population/literacy/SC-ST data for Ahmedabad.

Why: The 2011 Census used AMC's OLD ward delimitation — 57 core wards + 6 outgrowth areas (63 total, numbered 1–57 + 6 OG wards). AMC was re-delimited to the CURRENT 48-ward structure around 2021–2022 (Gujarat State Election Commission, to accommodate 50%-women-reservation with 4 councillors/ward). These are geographically DIFFERENT boundaries — Census "Ward 1" $\neq$ current "Ward 1 (Ramol Hathijan)". No official 57-to-48 crosswalk table exists publicly. Force-joining by Ward_ID would silently attach wrong-ward demographic data to the wrong current ward, producing a vulnerability score that looks real but is geographically meaningless — a serious, hard-to-detect integrity risk.

Source: Census of India NADA portal, dataset PC11_PCA-TV-2407 (verified: file contains real numbers — e.g. Census Ward 1 pop=49,408, Ward 57 pop=96,266 — but for the wrong ward geometry).

Prototype assumption: N/A — data source rejected outright rather than force-fit.

Future calibration: If AMC ever publishes ward-wise demographic data under the CURRENT 48-ward delimitation (e.g. from Census 2021+ once conducted, or an AMC health-department survey), it could be joined directly with no mismatch risk.

D8 — Vulnerability/exposure proxy: WorldPop population density + Sentinel-2 NDVI (via Google Earth Engine)

Decision: Replace the blocked Census approach with two GEE-derived, raster-based features that aggregate directly onto the CURRENT 48-ward polygons (no crosswalk needed):

1. `population_density` — from WorldPop/GP/100m/pop (2020), summed per ward then divided by ward area (km²).
2. `ndvi_green_cover` — mean NDVI from COPERNICUS/S2_SR_HARMONIZED (Sentinel-2), March–May 2024 composite (heat-season, cloud-filtered <20%).

Elderly%, outdoor-worker density, slum density, and healthcare access remain explicitly NULL — no standard, ward-matched public source exists for them in the available timeframe.

Why: Both datasets are RASTER (pixel-based), so they can be aggregated directly onto ANY polygon set, including the current 48-ward boundaries — completely sidestepping the D7 ward-delimitation problem. Population density is a recognized heat-vulnerability/exposure proxy (denser areas = harder evacuation, more urban heat island effect). NDVI is a recognized cooling-context proxy (less vegetation = less natural cooling, more heat retention).

Source: WorldPop Global Project 100m population (WorldPop/GP/100m/pop, verified real GEE catalog dataset); Sentinel-2 Surface Reflectance Harmonized (COPERNICUS/S2_SR_HARMONIZED).

Prototype assumption: WorldPop population is a MODELED estimate, not an official AMC ward population count — labeled as such wherever shown ("Population exposure estimated from WorldPop 100m gridded data, aggregated to ward polygons. Modeled estimate, not an official AMC count.").

Sanity check (passed):

- Total estimated population (48 wards) = 60.9 lakh vs Ahmedabad's actual estimated population (~63–65 lakh, current) — close ✓
- Total ward area = 441.8 km² vs AMC's official ~466 km² — close ✓
- NDVI range 0.09–0.31 — realistic for a dense urban city (no implausible outliers) ✓
- Spatial pattern matches known city geography: dense old-core wards (Bapunagar, Gomitpur, Amraiwadi, Saraspur-Rakhiyal) show highest population density; outer/newer wards (Maktampura, Thaltej, Jodhpur, Gota) show highest NDVI — consistent with real Ahmedabad urban structure ✓

Future calibration: Response capacity (healthcare access) still needs a real source — OpenStreetMap hospital/clinic locations + nearest-distance-per-ward is the planned next step (not yet built).

D9 — Priority Rubric v1: thermal-only (interim state, now superseded by v2)

Decision: Built and tested a rule-based priority rubric (`priority_rubric_v1()`) BEFORE vulnerability data existed, explicitly designed to never claim a Critical/High verdict it couldn't yet justify — it returned "Thermal-only: <tier> (partial – vulnerability/response capacity pending)" instead of fabricating a full verdict.

Why: Following the project's no-fabrication rule — better to show a clearly-labeled partial result than a fake-looking complete one. Also let the team validate the rubric's logic and CTBI-tier thresholds independently of the (at-the-time) blocked vulnerability data.

Source: `src/khushi_day3_sanity_and_priority.py`; tested on real ward-level CTBI from `ward_thermal_daily.csv`.

Note on ward-map "flatness" at this stage: Since `ward_thermal_daily.csv` attaches ONE city-wide CTBI value to every ward on a given date (CTBI.csv is a single time-series, not computed per grid-point/per-ward — a separate, documented Day 3 limitation), any single-date snapshot at this stage showed all 48 wards in the SAME thermal tier. This was correctly identified as expected behavior, not a bug (confirmed in Vrinda's Day 3

choropleth work, which showed real TIME variation — April→May→June — instead of misleadingly picking one date to imply false spatial variation).

Superseded by: D10 (below).

D10 — Priority Rubric v2: real vulnerability plugged in

Decision: Updated the rubric (`priority_rubric_v2()`) to use REAL vulnerability, now that D8's GEE data exists:

Vulnerability classification rule: A ward is "High vulnerability" if EITHER:

- `population_density` is in the top tercile ($\geq 33,055$ people/km², the busiest/hardest-to-evacuate wards), OR
- `ndvi_green_cover` is in the bottom tercile (≤ 0.1461 , least natural cooling)

Either condition alone is sufficient — they represent independent heat-vulnerability pathways (crowding vs. lack of green cooling).

Result: 23 of 48 wards classified High vulnerability, 25 Low — a real, spatially-varying split (dense old-core wards like Bapunagar, Gomtipur, Amraiwadi, Khadia, Danilimda; outer/leafier wards like Ghatlodia, New Wadaj, Bodakdev classified Low).

response_capacity is STILL not available (healthcare-access via OSM not yet built). So v2 introduces an honest intermediate state instead of a full verdict:

- "`<tier> (partial – response capacity pending)`" — uses REAL thermal (CTBI) + REAL vulnerability, but is explicitly capped BELOW "CRITICAL", since the locked rule (High thermal + High vulnerability + Low response capacity = CRITICAL) requires response capacity to be genuinely confirmed low, not assumed.

Why this design: Never claim CRITICAL without all three real factors — CRITICAL is the highest-stakes label on the dashboard (it's meant to trigger real interventions), so it gets the strictest evidence bar.

Source: `src/khushi_priority_rubric_v2.py`; tested on the full 48-ward × 546-day dataset (`priority_rubric_v2_full_test.csv`).

Future calibration: Once healthcare-access (OSM hospital distance) is built, response_capacity= can be passed as real "Low"/"High" instead of None, and the rubric will start returning genuine CRITICAL verdicts with no code changes — the function already supports it.

Open items (not yet resolved, tracked here so nothing is silently dropped)

Item	Status	Blocker
Response capacity (healthcare access)	● Not started	Needs OSM hospital/clinic query + nearest-distance-per-ward computation
NCTL/baseline night-window 2-hour mismatch (D5)	● Documented, not fixed	Low priority — doesn't change qualitative pattern
elderly_pct, outdoor_worker_density, slum_density	● Explicitly deferred to Future/Phase 2	No reliable ward-matched public source found in timeframe
Live SMS/WhatsApp alert delivery	● Not started (per playbook, API stub only)	Not a blocker — stub is sufficient per Day 5 plan